Thabani Chibanda

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PROFESSIONAL SUMMARY

Software Engineer & System Architect with a track record of driving distributed systems architecture, reactive data platforms, and high-performance backend systems across enterprise environments and ambitious technical ventures. Proven progression from leading core cloud and ML telemetry platforms at Amazon to architecting end-to-end domain-driven systems, custom distributed messaging engines, and low-latency data infrastructure as a Principal Architect. Specialist in functional programming (OCaml, Scala), type-safe system design, distributed consistency (Saga orchestration, server-side idempotency), and cost-optimized data analytics (DuckDB, AWS). Experienced in establishing engineering standards, authoring technical RFCs, and mentoring teams through complex technical execution.

TECHNICAL SKILLS

- **Languages:** Java, Python, TypeScript, Scala, OCaml, JavaScript
- **Cloud & Infrastructure:** AWS (DynamoDB, S3, Lambda, CloudFormation, CloudFront, EC2, Fargate, RDS), Docker, Linux
- **Databases & Search:** PostgreSQL, SQL, DuckDB, Elasticsearch, GraphQL
- **Web & Tools:** React, Express, HTML5/CSS3, HTMX, Git

PROFESSIONAL EXPERIENCE

pahukama.com | REMOTE
Lead Engineer & Principal System Architect | 08/2023 – Present

- System Architecture & DDD: Designed and authored a distributed modular monolith backend in OCaml utilizing Domain-Driven Design (DDD) principles, enforcing bounded contexts, type-safe domain models, and decoupled service boundaries.
- Distributed Reliability & Sagas: Engineered a custom Saga orchestration engine over an asynchronous messaging bus to execute distributed multi-step transactions with automated compensations and explicit idempotency guarantees.
- High-Throughput Analytics & Data Platform: Constructed a serverless analytics engine utilizing AWS (DynamoDB, S3, Lambda, CloudFront) and embedded DuckDB, executing zero-copy Parquet queries against S3 data lakes for low-latency CRM insights.
- Performance & UI Resilience: Integrated HTMX with server-side rendering (SSR) and token-based idempotency mechanisms to eliminate client-side state divergence and compress tail latencies across global user sessions.
- • Technical Leadership & Governance: Lead a team of 3 software engineers across system design RFCs, code architecture standards, and production releases, scaling the system to 1,000+ accounts and 500+ processed transactions in year one.

Tech Stack: OCaml/ReasonML, HTMX, React, Vue.js, DuckDB, AWS (EC2, Lambda, DynamoDB, S3, CloudFront, CloudFormation) React, AWS(EC2, Lambda, Dynamodb, S3), DuckDB

thatDot | Portland, OR
Software Engineer (Contract) | *01/2023 – 04/2023*

- **Architecture & Strategic Ownership**: Spearheaded the end-to-end technical strategy and system architecture for a real-time graph streaming data insights platform from conception to production MVP.
- **Cross-Functional System Design**: Led multi-team architecture reviews to define technical specs, data flow topologies, API contracts, and low-latency frontend/backend integration boundaries.
- **Stream & API Optimization**: Refactored core Scala backend services and streaming endpoints to deliver reactive, low-latency graph analytics into a React dashboard layer.

Tech Stack: Scala, TypeScript, React, Graph Data Systems, Streaming APIs

Amazon, Prime Video | Seattle, WA
Software Development Engineer | *01/2021 – 01/2023*

- **Distributed Platform Engineering**: Designed, scaled, and operated high-throughput distributed cloud services powering automated quality testing and telemetry across global Prime Video mobile applications.

- **Machine Learning & Triage Automation**: Built automated ML data ingestion pipelines for log clustering and anomaly triage, converting raw client telemetry into actionable real-time stability insights for iOS/Android engineering.
- **Cross-Org Telemetry Infrastructure**: Architected a unified analytics and diagnostic platform used across multiple engineering organizations, significantly compressing mean time to detection (MTTD) for client-side regressions.
- **Operational & Technical Ownership**: Served as primary service owner across the full technology stack—driving architecture reviews, operational excellence baselines, on-call rotation standards, and long-term capacity planning.

Tech Stack: Java, Python, TypeScript, React, AWS (EC2, Lambda, Fargate, Amplify, S3, DynamoDB, RDS, CloudFormation), PostgreSQL, Elasticsearch, Docker

Event Dynamic (now Tillion) | Dallas, TX
Full Stack Software Developer | *04/2020 – 12/2020*

- **Machine Learning API Integration**: Collaborated directly with Data Science teams to expose AI-driven dynamic pricing models as high-throughput, business-facing GraphQL and REST APIs.
- **Platform Overhaul & Performance**: Led the front-to-back refactoring of customer-facing dynamic pricing applications, optimizing API latency, client state management, and UI rendering performance.
- **MVP Delivery & System Lifecycle**: Driven end-to-end design, implementation, and deployment of dynamic web platforms within a fast-paced team, establishing testing and deployment standards across all lifecycle phases.

Tech Stack: Java, Scala, JavaScript, React, PostgreSQL, GraphQL, Docker, AWS (EC2, RDS)

Express Scripts | St. Louis, MO
Full Stack Developer (Contract) | 08/2019 – 03/2020

- **Enterprise Financial Platform Engineering**: Developed high-reliability contract analysis and financial evaluation microservices to support underwriting operations during large-scale enterprise acquisition integrations.
- **Data Integration & Legacy Migration**: Architected backend data integration pathways to seamlessly ingest and normalize multi-source client datasets into core relational databases.
- **Deployment & Release Management**: Coordinated production release cycles and automated CI/CD pipelines in Jenkins while building developer tooling and operational documentation to streamline internal workflows.

Tech Stack: Java, React, PostgreSQL, OracleDB, Jenkins

EDUCATION

Cook Systems FastTrack'D Developer Program | Memphis, TN
Accelerated Enterprise Software Engineering & Full-Stack Systems Training | 04/2019 – 07/2019